

Quantium Retail Analytics

Store Trial Evaluation & Customer Segmentation

Retail Analytics | Customer Segmentation | Quasi-Experimental Design | Statistical Testing

264,836

Transactions Analyzed

270

Stores

3

Trial Stores Evaluated

The Business Problem

TASK 1 — CUSTOMER UNDERSTANDING

Which customers drive chip revenue?

The category management team lacked a clear understanding of chip purchasing behavior across customer segments — making promotional and shelf strategy decisions unclear and unsupported by data.

TASK 2 — TRIAL EVALUATION

Did the new store layout work?

A new store format was trialed in 3 stores. Without a proper control group comparison, the business couldn't determine whether any sales change was driven by the layout or external factors.

AIM

Use transactional and customer data to identify the highest-value chip purchasing segments and deliver a statistically rigorous evaluation of the store trial — giving the category manager a data-driven basis for promotional strategy and rollout decisions.

Methodology

01 Data Cleaning & Preprocessing

Cleaned 264,836 chip transactions across 270 stores — removed outliers, standardized product names, extracted pack size and brand features

02 Customer Segmentation

Normalized purchase volume and spend by customer count to identify which lifestage and premium segments over-index on chip purchases

03 Brand Affinity Analysis

Calculated preference scores per segment to identify brand over-indexing relative to population average

04 Control Store Selection

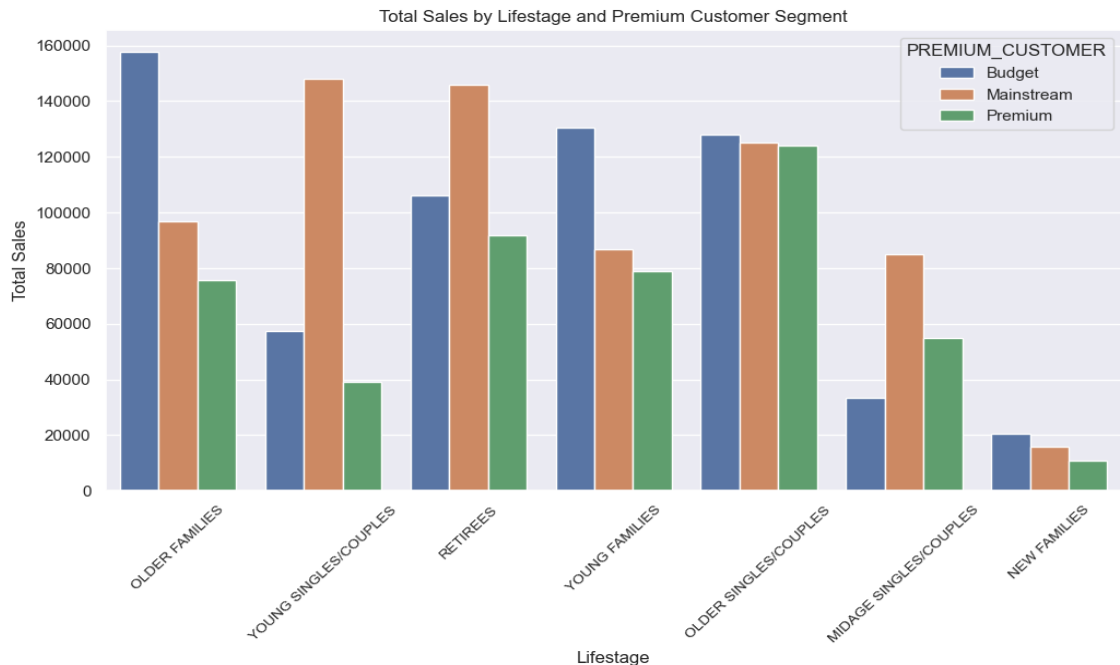
Each trial store paired with the most statistically similar non-trial store using Pearson Correlation on pre-trial monthly sales data

05 Trial Evaluation — 95% CI Testing

Compared trial vs control store performance across the trial period using confidence interval testing to determine statistical significance of uplift

Customer Segmentation — Who Buys Chips?

TOTAL SALES BY SEGMENT



KEY FINDINGS

Budget Older Families

Highest revenue segment — over-indexing on chip purchases relative to population share

Mainstream Young Singles

Second highest revenue driver with strong brand affinity across all chip brands

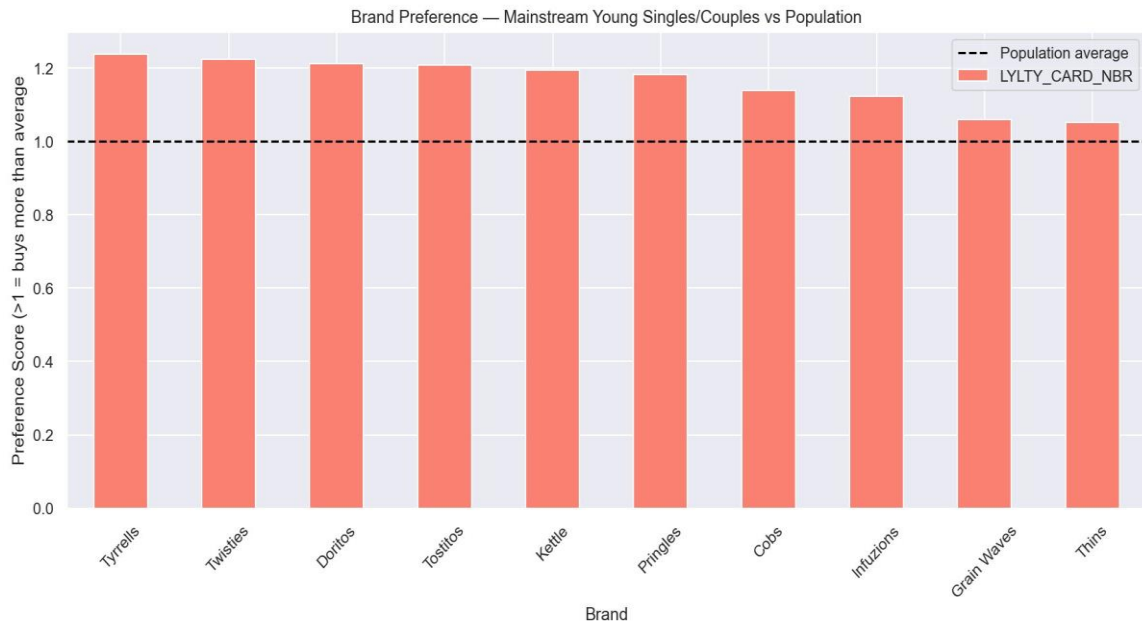
Spend per customer is equal

\$32-\$35 across top segments — total revenue difference driven by customer traffic, not individual spend

Recommendation: Prioritize shelf space, promotional investment, and new product ranging toward Budget Older Families and Mainstream Young Singles/Couples across all 270 stores.

Brand Affinity — Mainstream Young Singles/Couples

PREFERENCE SCORE vs POPULATION AVERAGE (score > 1 = over-indexes on that brand)



Tyrrells leads

Highest preference score among Mainstream Young Singles — 24% above population average

Twisties & Doritos

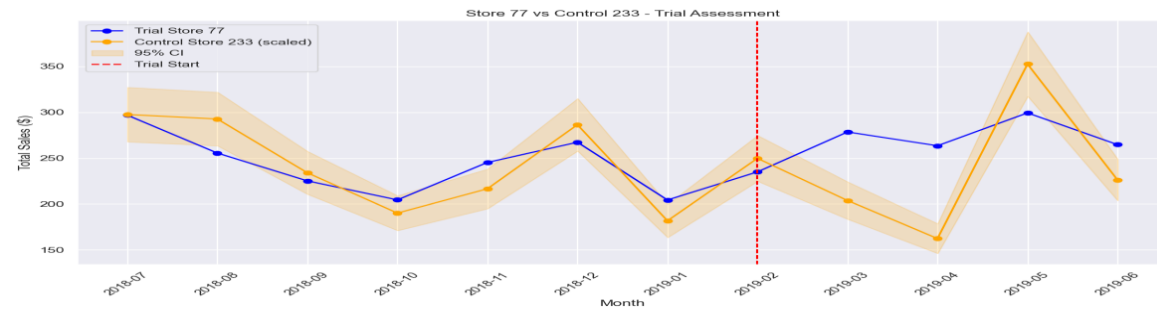
Second and third — segment over-indexes across all top 10 brands without exception

Promotional opportunity

Every brand scores above 1.0 — this segment is a reliable chip buyer regardless of brand, making it ideal for promotional targeting

Both Budget Older Families and Mainstream Young Singles/Couples over-index across most chip brands — supporting a broad promotional strategy rather than brand-specific targeting for these segments.

Store Trial Evaluation — Results

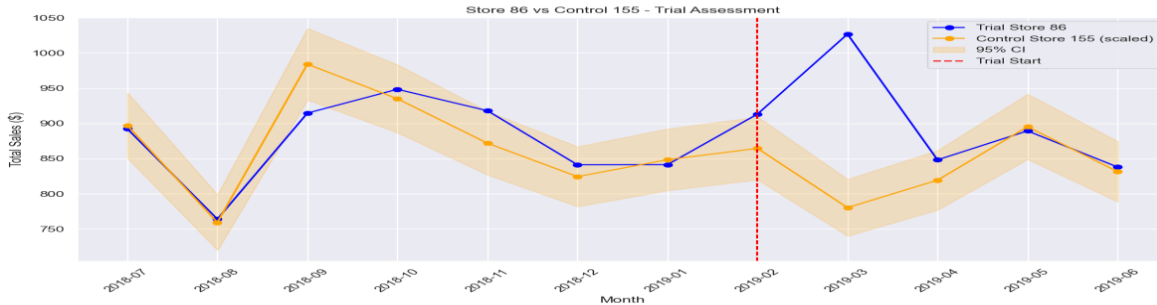


TRIAL vs CONTROL STORE PERFORMANCE — 95% CONFIDENCE INTERVAL TESTING

Store 77 ✓

Significant uplift

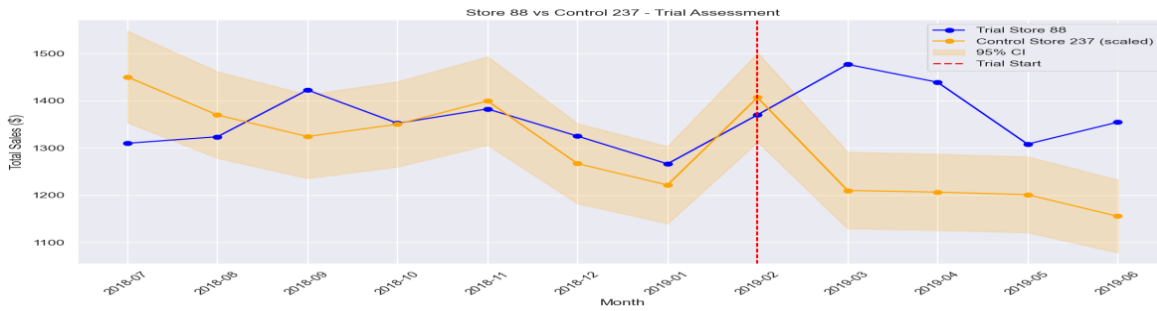
Sales exceeded 95% CI upper bound. Uplift driven by new customer acquisition during trial period.



Store 86 —

Short-lived uplift

Early uplift that couldn't be sustained. No negative impact. Layout had no harm — investigate local competition.



Store 88 ✓

Significant uplift

Strongest result. Driven by both 13% customer growth and 2–11% increase in per-customer transactions.

Recommendations

FOR THE CATEGORY MANAGEMENT TEAM

01 Prioritize Budget Older Families & Mainstream Young Singles

These two segments drive the highest chip revenue across all 270 stores. Shelf space, promotional campaigns, and new product ranging should be oriented toward their preferences — broad brand affinity means any chip promotion targeting these segments will be effective.

02 Roll out the new store layout — prioritize Store 77 & 88 profiles

2 of 3 trial stores showed statistically significant uplift at 95% confidence. The layout produces a positive effect on chip sales and should be deployed chain-wide. Prioritize stores that share the customer profile of Store 77 and Store 88 for maximum impact.

03 Investigate Store 86 before rollout in similar locations

Store 86 showed early uplift that did not sustain. The layout caused no harm but the short-lived effect may reflect local competitive dynamics. Additional in-store activation or promotional support may be required to sustain uplift in stores with similar profiles.

Next Steps & Limitations

NEXT STEPS

Expand the trial

A larger trial across more stores would provide more statistically robust conclusions before full chain rollout

Investigate Store 86

Understand whether the short-lived uplift is driven by local competition or layout execution — in-store observation recommended

Segment-specific promotions

Design targeted promotional campaigns for Budget Older Families and Mainstream Young Singles with measured uplift testing

LIMITATIONS

Trial size

Only 3 trial stores — statistical power is limited and conclusions should be validated with a larger sample

Causal inference

95% CI testing confirms uplift but cannot fully isolate layout effect from external factors — difference-in-differences would strengthen claims